

FoldShield⁺⁺ Structural Analysis Report

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Input Structures

Reference PDB	1UBQ.pdb
Query PDB	1J85.pdb
Mode	ca
Ref chain	A
Query chain	A

Classical Coordinate Metrics

These metrics saturate near 1.0 even when function diverges. Use FoldShield++ signals below for structural perturbation insight.

Metric	Value	Note
TM-score	0.300	>=0.50 same fold · >=0.90 nearly identical
RMSD (A)	4.000	Lower is better · >3.0 A = significant deviation

FoldShield++ Signal Scores — Phase 1 + Phase 2

All active signals with scores, fusion weights, and interpretation. Phase 2 signals are highlighted in blue.

Braid similarity	0.381	0.20	weak braid similarity
Motif similarity (UL-DSL)	0.517	0.37	moderate motif similarity
SSEF entropy (Phase 2) [P2]	0.679	0.20	moderate grammar rewrite — conformational transition regime
Persistent homology (Phase 3) [P2]	0.526	0.23	partial persistence similarity

Fusion Layer v2 — Combined Score

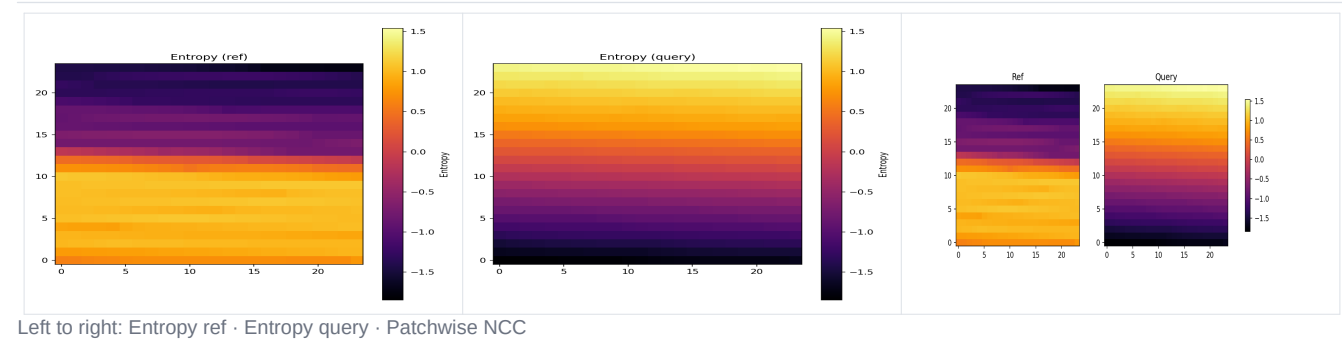
0.5237	phase2	weak structural similarity

Phase 2 — Perturbation Classification

SSEF score	Local topology	Gap score	Perturbation class
0.679	0.046	0.633	Conformational transition

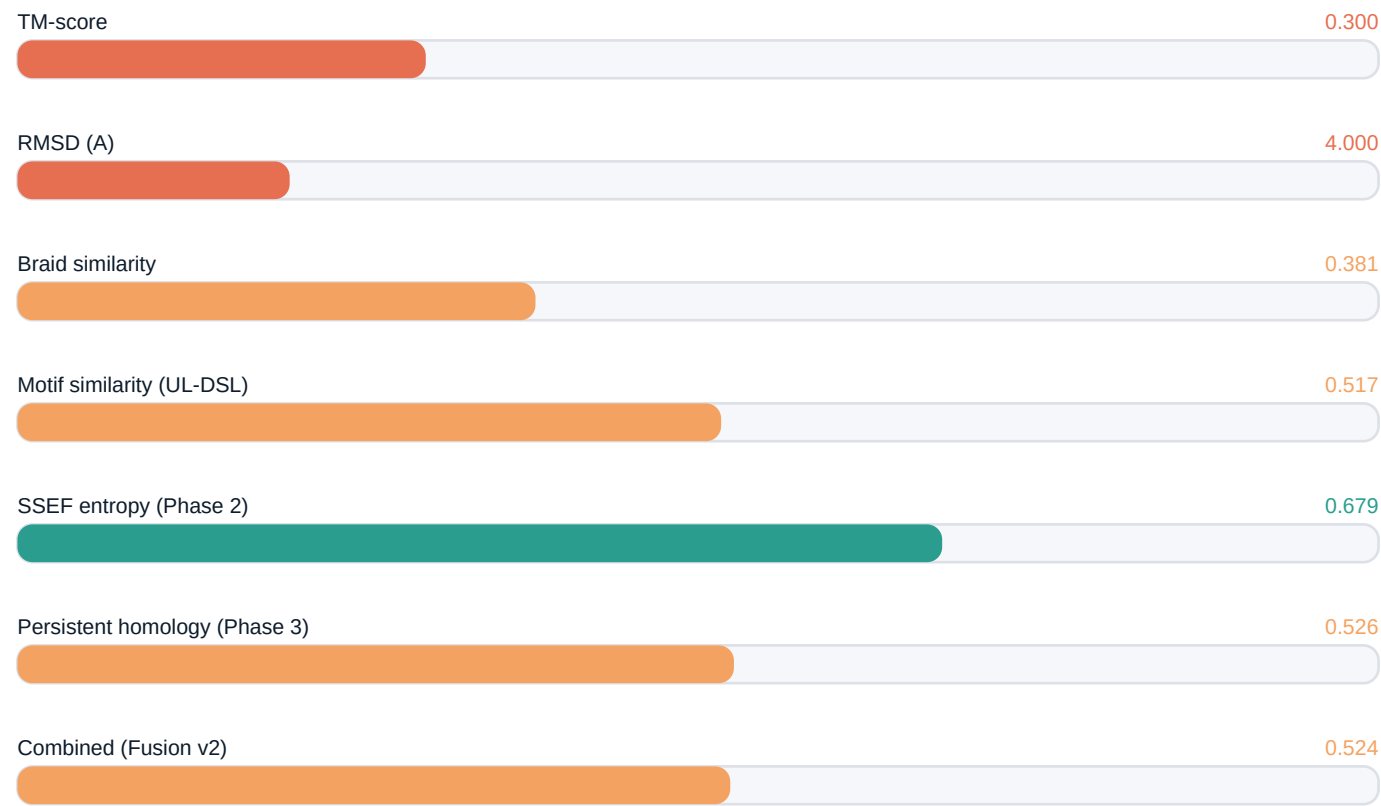
Gap = SSEF minus local topology. <0.15 buried-core mutation | 0.15-0.36 fold-switch | >0.36 conformational transition

Entropy Maps & Signal Profiles



Signal Score Bars

Visual 0-1 scale for all active signals. Green ≥ 0.66 / Amber 0.33-0.66 / Red < 0.33 .



Symbolic Score Detail

Symbolic Similarity Score	0.381	weak similarity
Behavioural Correlation Similarity	0.260	weak entropy similarity
Structural Topology Similarity	0.484	partial generator overlap
Structural Topology Ordering Similarity	0.498	moderate pattern similarity

FoldShield++ Phase 2 | RexCrux Research Laboratory | Signals: Braid / UL-DSL / Global Topology / Local Topology / SSEF Entropy | Fusion: Ridge Regression weighted linear combination

Signal Analysis — Figures

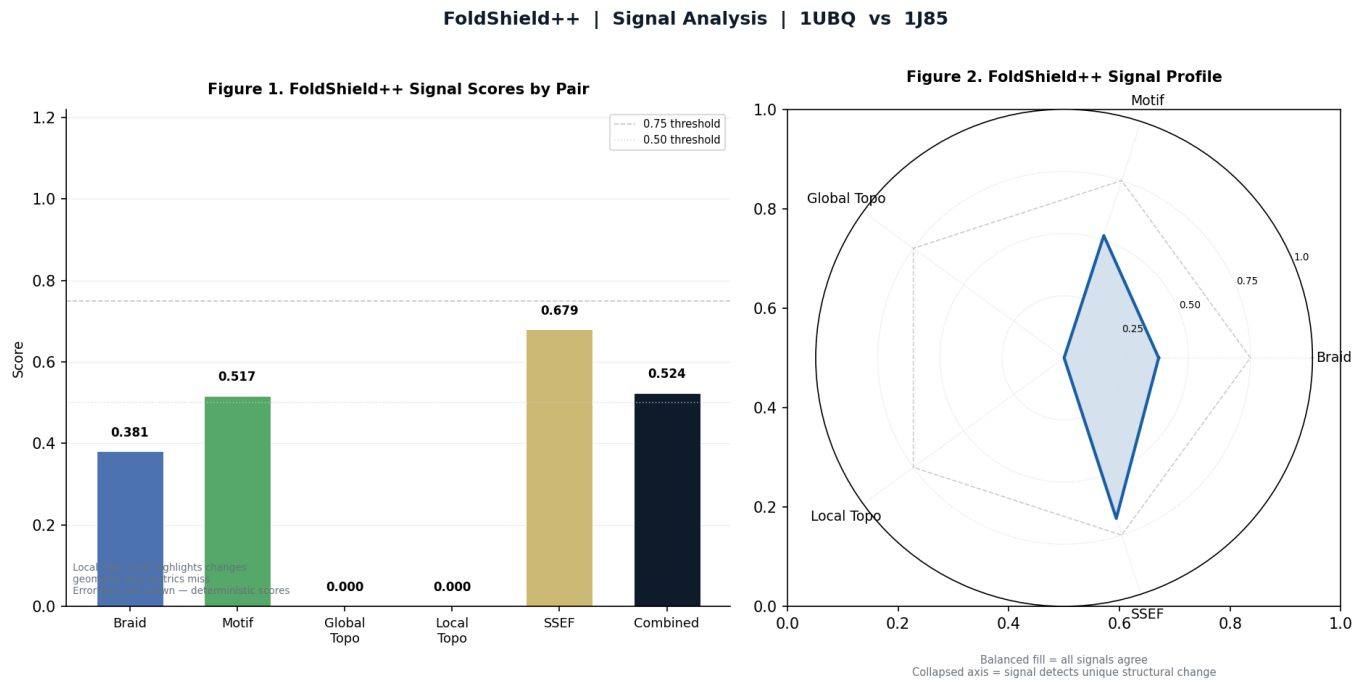


Figure 1: Bar chart comparing all five signals and the Combined score. Local Topo (red) highlights structural changes geometry-only metrics miss. Dashed = 0.75 · Dotted = 0.50. Error bars not shown — scores are deterministic. Figure 2: Radar chart showing the five-signal profile. Balanced fill = all signals agree. Collapsed axis = signal detects unique structural change.